

CS 217/240: Artificial Intelligence (AI) and Machine Learning (ML)

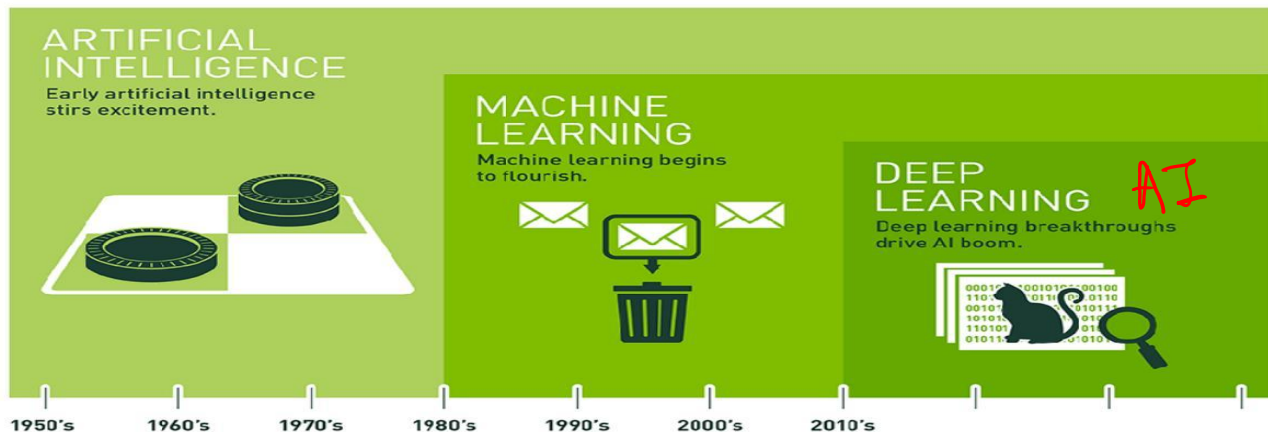
Sunita Sarawagi
CSE

sunita@iitb.ac.in

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The field of AI and ML

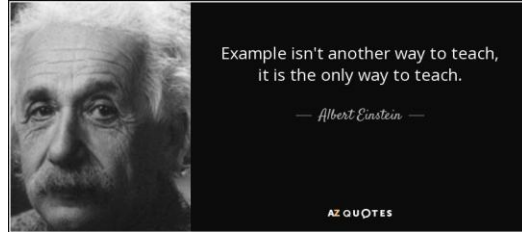
- AI: An agent (software) that can perceive the environment, ‘think’ about it, understand the human-provided goal, and take action
- ML: A sub-field of AI that focuses on how to improve the action by ‘learning’ from experience (also called data)



Machine Learning

- Gives a computer the ability to learn to perform tasks without explicit programming
 - Tasks: labeling images, recognizing faces, converting voice to text, forecasting, translation, editing grammar,...
- Learn how?
 - Collect examples of (input, output) pairs
 - Train a model or a function to fit the examples
- Perform tasks how?
 - Given a new input, invoke the model and reads its output

Why learn from examples instead of writing programs?



Difficult for a programmer to express perfect input-output relationship but easy for humans to label, example:

- Labeling an audio sample as “Alexa” is easier than writing programs
- Humans can easily recognize a person given an image but cannot write program for automatically spotting people in any image.

Core Task of ML: Classification/Regression

Input: any form

x



y

Predicted output: a single variable

- Real-valued: regression
- Categorical: classification

Learn

Training data containing several correct (x , y) pairs

Input: Image (see)



→ Panther



→ Dog

Surveillance



Glaucoma percent

→ 1%



→ 70%

Health

Input: text (comprehend)

→ Koi kuda nahi utatha

→ Garbage

→ Garbage collectors asking for money

→ Ethics

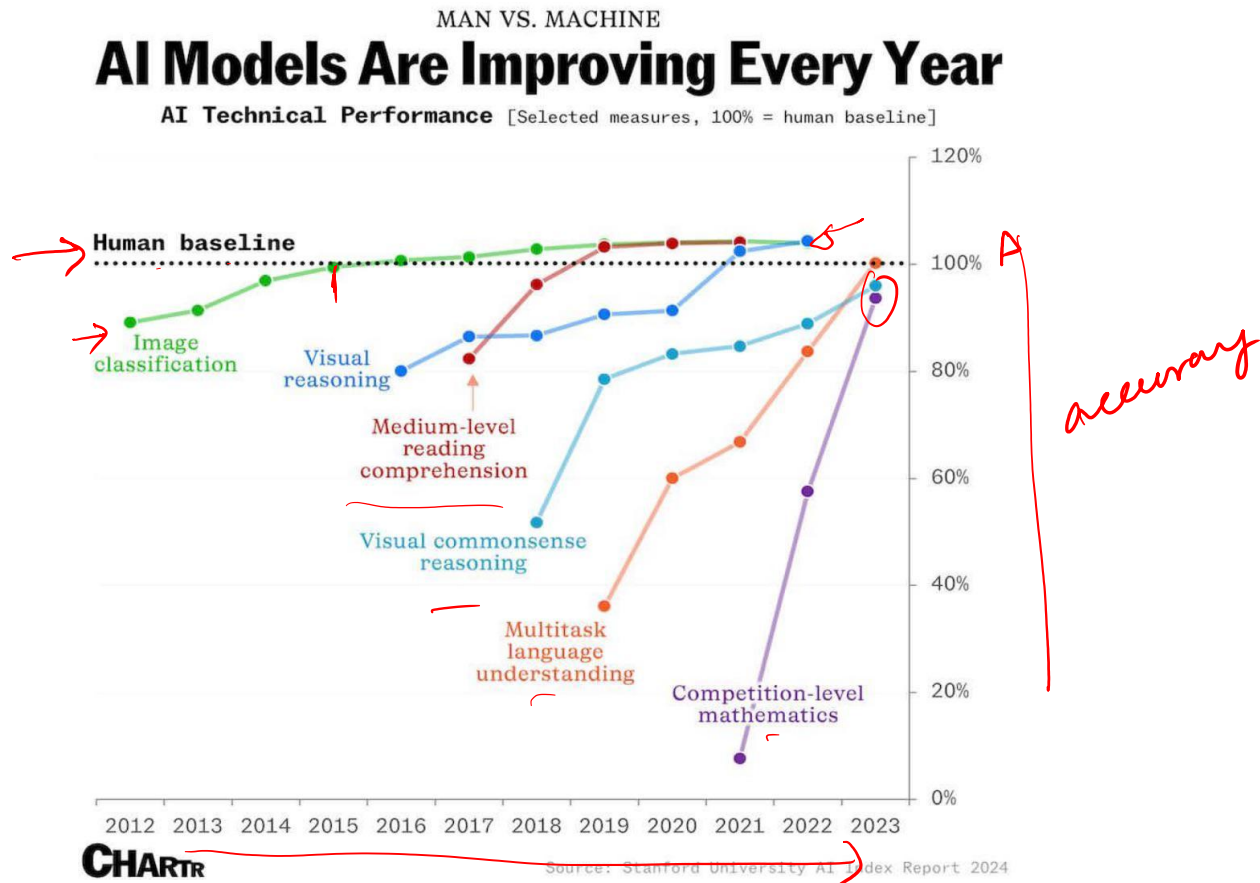
Complaint management system

And thousands of others: forecasting, recommendation, advertising, science, languages, vision.....

Innumerable Applications

- Ad placement in search engines
- Inventory management: Predict sale of soft drinks in outlets based on weather, events (sport)
- Scheduling: predicting traffic, flight arrival times.
- Fraud detection: telecommunications, financial transactions
 - from an online stream of event identify fraudulent events
- Banking: loan/credit card approval 
 - predict good customers based on old customers
- Customer relationship management:
 - identify those who are likely to leave for a competitor.
- Targeted marketing:
 - Recommendation of Movies, Books, Products on E-commerce sites

AI/ML seeing unprecedented growth..

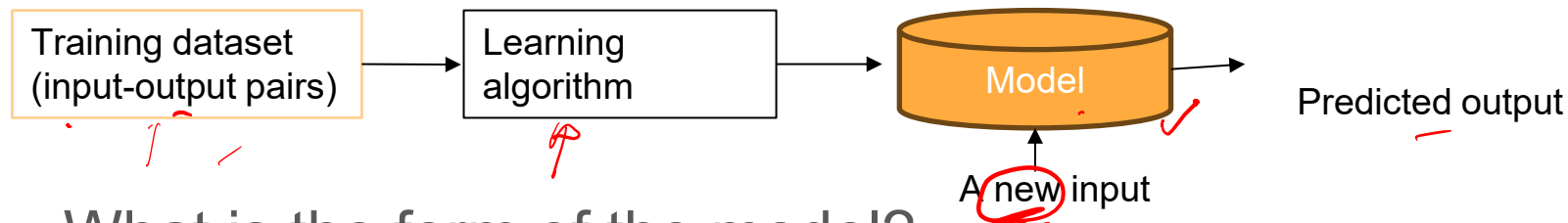


Scope of the course

This course is not about using ML to solve cool tasks but about understanding the underlying *foundational principles* and *methodologies* behind learning from examples so that

- You can teach machines to solve more such tasks using your knowledge of the founding principles
- You can intelligently drive the design and training of ML models
- Debug non-performance

Questions that we will address in this course



- What is the form of the model?
- How does the learning algorithm work?
- How do we make ensure generalization so the model will produce (mostly) correct outputs on test/deployment data?
- How to tractably handle complex inputs and outputs?
- What if we do not have enough labelled examples for each task? Can we share learning across tasks?

Course Information

[Follow this page](#)

Study material

- The most relevant text books mentioned on the course webpage
- Each topic will contain pointer to reading material on that topic
- My slides and/or board work on the tablet will be available for reference.
- Exams will be open notes, but you cannot xerox someone else's notes or my board work.

Academic Integrity

Abide by an honour code and do not be involved in any plagiarism. If caught for copying or plagiarism, name of both parties will be handed over to DDAC
For all graded evaluations

- Write what you know in your own words
- If you refer to external material, including LLMs cite

The study process

- Pay attention to the class: don't hesitate to ask questions
 - (Old Chinese proverb)
 - The one who asks a question is stupid only once, the one who does not ask questions is stupid forever.
- For every 1.5 hours of lectures, spend at least 1.5 hours in revising the lecture after class
- Do the homework yourself
- If you have difficulty, seek help early on in the semester

How to get help?

- Ask questions in class
- Ask questions on discussion forums (piazza)
- Attend help-sessions
- Come to office hours of instructor

- Be Calm
- Be Organized
- Enjoy the process..